

3.1.11 Electrode potentials and electrochemical cells (A-level only)

Redox reactions take place in electrochemical cells where electrons are transferred from the reducing agent to the oxidising agent indirectly via an external circuit. A potential difference is created that can drive an electric current to do work. Electrochemical cells have very important commercial applications as a portable supply of electricity to power electronic devices such as mobile phones, tablets and laptops. On a larger scale, they can provide energy to power a vehicle.

3.1.11.1 Electrode potentials and cells (A-level only)

Content	Opportunities for skills development
IUPAC convention for writing half-equations for electrode reactions. The conventional representation of cells. Cells are used to measure electrode potentials by reference to the standard hydrogen electrode. The importance of the conditions when measuring the electrode potential, E (Nernst equation not required). Standard electrode potential, $E^\ominus$, refers to conditions of 298 K, 100 kPa and 1.00 mol dm⁻³ solution of ions. Standard electrode potentials can be listed as an electrochemical series. Students should be able to: • use $E^\ominus$ values to predict the direction of simple redox reactions • calculate the EMF of a cell • write and apply the conventional representation of a cell.	AT j and k PS 1.1 Students could make simple cells and use them to measure unknown electrode potentials. AT a, b, j and k PS 2.1 and 2.4 Students could be asked to plan and carry out an experiment to investigate the effect of changing conditions, such as concentration or temperature, in a voltaic cell such as Zn Zn²⁺ Cu²⁺ Cu AT j and k PS 2.2 Students could use $E^\ominus$ values to predict the direction of simple redox reactions, then test these predictions by simple test-tube reactions.
Required practical 8 Measuring the EMF of an electrochemical cell.	

3.1.11.2 Commercial applications of electrochemical cells (A-level only)

Content	Opportunities for skills development
Electrochemical cells can be used as a commercial source of electrical energy. The simplified electrode reactions in a lithium cell: Positive electrode: $\text{Li}^+ + \text{CoO}_2 + \text{e}^- \rightarrow \text{Li}^+[\text{CoO}_2]^-$ Negative electrode: $\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$ Cells can be non-rechargeable (irreversible), rechargeable or fuel cells. Fuel cells are used to generate an electric current and do not need to be electrically recharged. The electrode reactions in an alkaline hydrogen–oxygen fuel cell. The benefits and risks to society associated with using these cells. Students should be able to: • use given electrode data to deduce the reactions occurring in non-rechargeable and rechargeable cells • deduce the EMF of a cell • explain how the electrode reactions can be used to generate an electric current.	Research opportunity Students could investigate how knowledge and understanding of electrochemical cells has evolved from the first voltaic battery.